



NATIONAL TEXTILE UNIVERSITY

Department of Computer Science

Lab # 5: Programming Fundamentals (CSC-1071)

Basic Information	
Tools:	Dev-C++ 5.4.1
Objectives	1. do-while Loops 2. Sentinel Controlled Loops
Note	Solve the following problems using the concepts we have covered so far

Sentinel Controlled Loops and Do-While Loops

1. Display numbers from 1 to 20 with **while**, **do-while** and **for loop**. Start value of loop is 0 , ending is less than 20 and sequence is plus 1. Observe the behavior
2. Input the marks of ten students and display their average, minimum and maximum. Use **do-while**. **Following is sample output:**

```
Enter 10 Programming marks:23 78 56 89 98 45 67 76 67 89

Average=68.8
Minimum Marks=23
Maximum Marks=98
-----
```

3. Write a program that keeps asking the user to enter a number until the user enters -1, which is the sentinel value. Output all entered numbers.
4. Write a program that keeps asking the user to enter positive numbers until a negative number is entered. The negative number should not be printed.
5. Write a program that asks the user to input marks. The program should stop when the user enters -1 and calculate the average of the marks. Also show maximum and minimum marks

```
Enter any length of numbers(press -1 to end):45 46 47 48 -1

Average=46
Minimum Marks=45
Maximum Marks=48
-----
```

6. Write a program that asks for the user's password until they input "12345" as the correct password.
7. Write a program that asks for words until the word "exit" is entered. The program should then output how many words were entered.
8. Ask user to enter an integer number of any size and display its reverse and check whether it is a Palindrome or Not
9. Ask user to enter an integer number of any size and display its reverse and addition. **(Solve with both ways of Sentinel Control). Sample output is given below:**

```

Enter Integer number of any size:1234

Reverse is:4321
Addition=10
-----

```

10. Display largest integer and smallest among the several integers entered by the user. Uses **Y** OR **y** as sentinel value
11. In Fibonacci series next number is calculated by adding previous two numbers. **0, 1, 1, 2, 3, 5, 8, 13...** Ask the user to enter the bound and show Fibonacci numbers up to that bound. For example, if user enters 60 then your program will show. **0, 1, 1, 2, 3, 5, 8, 13, 21, 35, 56**

Nested Loops

1. Input the gender of 10 students in the form of 'm, M' or 'f, F'. Force user to enter on m,M, f, F. Display how many students are male and female. Use **do-while** and **for** loop
2. Display following Multiplication table with loops only:

Multiplication Table									
	1	2	3	4	5	6	7	8	9
1	1	2	3	4	5	6	7	8	9
2	2	4	6	8	10	12	14	16	18
3	3	6	9	12	15	18	21	24	27
4	4	8	12	16	20	24	28	32	36
5	5	10	15	20	25	30	35	40	45
6	6	12	18	24	30	36	42	48	54
7	7	14	21	28	35	42	49	56	63
8	8	16	24	32	40	48	56	64	72
9	9	18	27	36	45	54	63	72	81

3. Output following Pattern


```

1
1 2
1 2 3
1 2 3 4
1 2 3 4 5

```
4. Display above pattern by inputting the ending value from user
5. Display following Pattern:


```

5 4 3 2 1
4 3 2 1
3 2 1
2 1
1

```
6. Half Pyramid of Numbers:


```

1
2 2
3 3 3
4 4 4 4
5 5 5 5 5

```
7. Print a Square Matrix of n x n with stars *
8. Try following shapes in separate programs and then try them in one program:


```

*      *****      *****      *
      **      ****      ****      **

```


15. Pyramid Pattern with Numbers:

```
1
2 3
4 5 6
7 8 9 10
```

16. Display the following pattern using the nested loops:

```
1
2  4
3  6  9
4  8  12 16
5  10 15 20 25
```